

PEMS05

Component	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	10.39±0.06	11.00±0.06	12.25±0.10	11.10±0.06
+ FiLM calibrator (warm-up only)	10.42±0.02	10.99±0.03	12.09±0.05	11.06±0.03
+ online delayed-label TTA (All)	9.84±0.03	10.45±0.04	11.61±0.06	10.52±0.04
+ local clone (Ours)	9.80±0.03	10.40±0.04	11.57±0.06	10.48±0.04

PEMS06

Component	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	13.10±0.21	13.91±0.19	15.54±0.23	14.02±0.19
+ FiLM calibrator (warm-up only)	14.01±0.17	14.77±0.21	16.25±0.26	14.86±0.20
+ online delayed-label TTA (All)	12.24±0.09	13.07±0.12	14.68±0.17	13.16±0.12
+ local clone (Ours)	12.15±0.09	12.97±0.12	14.55±0.16	13.06±0.11

Ablation-B: Calibrator Architecture (online TTA, `tta\_all`) — Avg MAE + #Params

Architecture	#Params	PEMS05	PEMS06
residual (raw-input MLP residual)	7821	10.75	13.65
norm (EMA scale-fix MLP residual)	7821	10.63	13.79
affine (static channel-wise $1+\gamma, \beta$)	24	10.89	14.15
lowrank (linear low-rank residual, r8)	460	10.80	13.87
adapter (bottleneck residual, r8)	468	10.68	13.80
gated_adapter (adapter + tanh gate, r8)	480	10.96	13.99
temporal_conv (depthwise conv over horizon)	4	10.86	14.14
film (Ours, conditional $\gamma\odot y+\beta$)	8601	10.52	13.16
adapter (parameter-matched, r150)	8562	10.58	13.26

Ablation-C: Calibrator as an Online-Adaptation Interface (warm-up only) — Avg MAE

Architecture	PEMS05	PEMS06
backbone / source (no calibrator)	11.10	14.02
film	11.06	14.86
affine	11.25	14.73
adapter	11.21	14.80
gated_adapter	11.30	14.51

Ablation-D: Knockout (leave-one-out from full Ours) — multi-backbone

Knockout row	`--method`	`--calibrator_arch`	removes
Ours (full)	`tta_ctx_local`	`film` nothing	(frozen backbone + FiLM + online TTA + local clone)
– local clone	`tta_all`	`film` the per-batch discardable local clone	(global calibrator still a)
– FiLM → affine	`tta_ctx_local`	`affine` FiLM's conditioning	net — **replaced** by a static $(1+\gamma)\cdot y + \beta$
– online TTA	`calibrator`	`film` online adaptation	— FiLM is warm-up-trained then **froze
– calibration stack → backbone	`backbone`	— the whole calibrator	→ falls back to the frozen backbone (ba

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	11.52±0.02	12.48±0.04	14.29±0.07	12.59±0.04	—
– local clone	11.58±0.02	12.54±0.04	14.34±0.06	12.65±0.04	+0.06
– FiLM → affine (naive)	12.44±0.05	13.41±0.07	15.28±0.10	13.54±0.07	+0.95
– online TTA (FiLM warm-up on	13.30±0.05	14.13±0.06	15.70±0.08	14.23±0.06	+1.64
– calibration stack → Online-A	12.76±0.04	13.68±0.04	15.46±0.08	13.80±0.04	+1.21

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	17.16 $\pm$ 0.05	18.72 $\pm$ 0.06	21.72 $\pm$ 0.12	18.93$\pm$0.07	—
– local clone	17.29 $\pm$ 0.05	18.83 $\pm$ 0.06	21.79 $\pm$ 0.11	19.04 $\pm$ 0.06	+0.11
– FiLM $\rightarrow$ affine (naive)	18.67 $\pm$ 0.09	20.21 $\pm$ 0.12	23.24 $\pm$ 0.22	20.43 $\pm$ 0.14	+1.50
– online TTA (FiLM warm-up on	20.32 $\pm$ 0.05	21.59 $\pm$ 0.07	24.09 $\pm$ 0.11	21.77 $\pm$ 0.07	+2.84
– calibration stack $\rightarrow$ Online-A	19.34 $\pm$ 0.19	20.78 $\pm$ 0.23	23.68 $\pm$ 0.30	21.01 $\pm$ 0.23	+2.08

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	9.80±0.03	10.40±0.04	11.57±0.06	10.48±0.04	—
– local clone	9.84±0.03	10.45±0.04	11.61±0.06	10.52±0.04	+0.04
– FiLM → affine (naive)	10.14±0.05	10.78±0.06	12.06±0.09	10.88±0.06	+0.40
– online TTA (FiLM warm-up on	10.42±0.02	10.99±0.03	12.09±0.05	11.06±0.03	+0.58
– calibration stack → Online-A	10.39±0.06	11.00±0.06	12.25±0.10	11.10±0.06	+0.62

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	12.15 $\pm$ 0.09	12.97 $\pm$ 0.12	14.55 $\pm$ 0.16	13.06$\pm$0.11	—
– local clone	12.24 $\pm$ 0.09	13.07 $\pm$ 0.12	14.68 $\pm$ 0.17	13.16 $\pm$ 0.12	+0.10
– FiLM $\rightarrow$ affine (naive)	13.17 $\pm$ 0.15	14.01 $\pm$ 0.13	15.66 $\pm$ 0.18	14.12 $\pm$ 0.13	+1.06
– online TTA (FiLM warm-up on	14.01 $\pm$ 0.17	14.77 $\pm$ 0.21	16.25 $\pm$ 0.26	14.86 $\pm$ 0.20	+1.80
– calibration stack $\rightarrow$ Online-A	13.10 $\pm$ 0.21	13.91 $\pm$ 0.19	15.54 $\pm$ 0.23	14.02 $\pm$ 0.19	+0.96

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	10.78±0.05	11.29±0.05	12.14±0.07	11.31±0.05	—
– local clone	10.83±0.06	11.33±0.06	12.17±0.08	11.35±0.06	+0.04
– FiLM → affine (naive)	11.52±0.10	11.97±0.10	12.77±0.13	12.01±0.11	+0.69
– online TTA (FiLM warm-up on	12.39±0.16	12.79±0.15	13.48±0.15	12.82±0.15	+1.50
– calibration stack → STAEform	11.81±0.10	12.22±0.09	12.98±0.12	12.26±0.10	+0.95

PEMS04

Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	16.17 $\pm$ 0.06	17.00 $\pm$ 0.10	18.42 $\pm$ 0.19	17.05$\pm$0.11	—
– local clone	16.26 $\pm$ 0.06	17.07 $\pm$ 0.09	18.48 $\pm$ 0.18	17.13 $\pm$ 0.10	+0.08
– FiLM $\rightarrow$ affine (naive)	18.08 $\pm$ 0.94	18.83 $\pm$ 0.92	20.16 $\pm$ 0.94	18.89 $\pm$ 0.93	+1.84
– online TTA (FiLM warm-up on	18.86 $\pm$ 0.17	19.49 $\pm$ 0.17	20.62 $\pm$ 0.22	19.54 $\pm$ 0.18	+2.49
– calibration stack $\rightarrow$ STAEform	18.66 $\pm$ 1.01	19.35 $\pm$ 0.97	20.60 $\pm$ 0.99	19.41 $\pm$ 0.98	+2.35

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	9.31±0.02	9.61±0.03	10.12±0.04	9.63±0.03	—
– local clone	9.34±0.02	9.63±0.03	10.14±0.04	9.65±0.03	+0.02
– FiLM → affine (naive)	9.64±0.05	9.90±0.05	10.40±0.07	9.93±0.05	+0.30
– online TTA (FiLM warm-up on	9.85±0.04	10.12±0.05	10.59±0.07	10.14±0.05	+0.51
– calibration stack → STAEform	9.88±0.06	10.12±0.06	10.60±0.07	10.16±0.06	+0.53

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Knockout	MAE@3	MAE@6	MAE@12	Avg	Δ Avg
Ours (full)	11.82±0.40	12.24±0.41	12.97±0.42	12.27±0.41	—
– local clone	12.24±0.53	12.65±0.54	13.37±0.56	12.68±0.54	+0.41
– FiLM → affine (naive)	12.39±0.40	12.76±0.39	13.45±0.38	12.80±0.39	+0.53
– online TTA (FiLM warm-up on	13.11±0.29	13.48±0.29	14.14±0.28	13.51±0.29	+1.24
– calibration stack → STAEform	12.55±0.45	12.89±0.45	13.56±0.43	12.94±0.44	+0.67

Ablation-1: Component Ablation — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	19.34±0.19	20.78±0.23	23.68±0.30	21.01±0.23
+ Residual Calibrator (warm-up, no TTA)	20.55±0.10	21.91±0.13	24.66±0.20	22.13±0.14
+ Delayed-label TTA (Ours)	18.23±0.12	19.72±0.11	22.75±0.19	19.97±0.13

Ablation-1: Component Ablation — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	10.39±0.06	11.00±0.06	12.25±0.10	11.10±0.06
+ Residual Calibrator (warm-up, no TTA)	10.52±0.02	11.13±0.04	12.35±0.07	11.22±0.04
+ Delayed-label TTA (Ours)	10.02±0.03	10.66±0.04	11.93±0.08	10.75±0.05

Ablation-1: Component Ablation — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	16.23±0.25	17.53±0.28	19.94±0.32	17.67±0.28
+ Residual Calibrator (warm-up, no TTA)	16.44±0.21	17.72±0.23	20.06±0.26	17.85±0.23
+ Delayed-label TTA (Ours)	14.77±0.14	16.21±0.16	18.86±0.18	16.36±0.16

Ablation-1: Component Ablation — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
Backbone (frozen)	12.76±0.09	13.80±0.09	15.75±0.10	13.92±0.09
+ Residual Calibrator (warm-up, no TTA)	12.94±0.06	13.96±0.06	15.86±0.08	14.08±0.07
+ Delayed-label TTA (Ours)	12.27±0.05	13.36±0.06	15.36±0.08	13.48±0.06

Ablation-2: Regularisers — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
no reg	10.08±0.03	10.72±0.04	12.00±0.07	10.82±0.04
prox only (λ_r)	10.09±0.03	10.73±0.04	12.01±0.08	10.82±0.04
cons only ($\lambda_c=.05$)	10.10±0.04	10.73±0.05	12.01±0.08	10.83±0.05
both (default)	10.08±0.04	10.72±0.05	12.00±0.09	10.81±0.05
cons only ($\lambda_c=.10$)	10.09±0.04	10.73±0.05	12.00±0.08	10.82±0.05
$\lambda_c=.10$ + prox	10.09±0.03	10.73±0.04	12.01±0.08	10.83±0.04

Ablation-2: Regularisers — MAE

Variant	MAE@3	MAE@6	MAE@12	Avg
no reg	12.35±0.05	13.43±0.06	15.45±0.08	13.55±0.06
prox only (λ_r)	12.34±0.05	13.42±0.06	15.45±0.08	13.55±0.06
cons only ($\lambda_c=.05$)	12.35±0.05	13.43±0.06	15.44±0.08	13.55±0.06
both (default)	12.36±0.07	13.44±0.08	15.45±0.09	13.56±0.08
cons only ($\lambda_c=.10$)	12.36±0.05	13.44±0.06	15.45±0.08	13.57±0.06
$\lambda_c=.10$ + prox	12.36±0.05	13.44±0.06	15.45±0.08	13.56±0.06